

Population-level spectral analysis of leaky integrate-and-fire neuronal archetypes

v2 — written for an FCS-native audience

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Abstract. The neuronal archetypes of De Maria et al. [1] are small LI&F network motifs whose qualitative behavioural primitives — contralateral inhibition produces winner-take-all selection, activator–inhibitor loops oscillate, positive loops are bistable — are properties of the topology, not the integer synaptic weights. The FCS approach charts these behaviours by Lustre encoding plus model checking with Kind2, on integer-grid sweeps. The discrete LI&F dynamics are hybrid (continuous subthreshold integration interleaved with instantaneous spike resets), which obstructs direct application of classical bifurcation tools to map the behavioural regimes analytically. This note takes the archetypes up one level of description: a population of LI&F neurons with distributed thresholds averages into a smooth Wilson–Cowan rate equation [2] governed by a sigmoidal gain function, and the classical toolkit (Jacobians, eigenvalues, bifurcation theory, pole placement) becomes immediately usable. We rebuild the contralateral archetype in this framework and show that the winner-take-all mechanism falls out of two lines of algebra: mutual inhibition is a feedback loop with gain $w_{12}w_{21}g_1g_2$, and the symmetry-breaking pitchfork is the unity contour of that loop gain — the same Barkhausen criterion used in classical electronics. Three hypotheses on the population framework are then tested numerically: (A) the spectral gap predicts the winner-take-all boundary at 99.96% classification accuracy; (B) the pitchfork and Hopf loci are derivable in closed form; (C) target oscillation frequencies are achievable by linear pole placement. The framework generalises to series, parallel, and positive-loop archetypes. Cross-validation against the discrete LI&F simulator agrees at the symmetric corner and reveals a second, spike-timing-driven bistability that the population framework cannot see — sharpening the scope of behaviours the framework can address. The population framework is offered as a **companion** to the FCS programme, not a replacement: it predicts the shape of behavioural parameter regions, where Lustre/Kind2 verifies properties at specific points.

1. Introduction

1.1. Why the lift

The FCS programme [1] verifies temporal-logic properties of LI&F archetypes by encoding them as Lustre nodes and discharging the properties to model checkers (Kind2) and theorem provers (Coq). The verification is point-wise: each cell of an integer weight grid is either certified (the property holds) or refuted (with a Lustre counter-example trace). The approach is rigorous, automatic, and delivers the kind of formal guarantee that makes archetypes a practical engineering primitive.

Two questions, however, lie just outside its natural reach. First, **the shape of the parameter region**. FCS Fig. 10 shows that contralateral inhibition’s winner-take-all stabilises within four ticks once the synaptic weights are large enough; what is the analytical form of that boundary,

and how does it scale beyond the sweep range $|w^{\text{LIF}}| \leq 40$ on which Fig. 10 is plotted? Second, **inverse design**. Given a target oscillation frequency, what synaptic weights realise it? The FCS apparatus can verify a candidate weight assignment but does not solve for one.

The obstruction to addressing both questions with classical methods is the LI&F reset rule itself. The membrane potential integrates linearly until it crosses threshold, then resets instantaneously to zero. This non-smoothness rules out Jacobians, spectra, pole placement, and bifurcation diagrams in any direct sense, because those tools all presuppose a differentiable vector field. The qualitative dynamical structure of LI&F archetypes has therefore been charted by exhaustive simulation or model checking, rather than by the closed-form methods available for smooth neural models [3], [4].

The classical remedy is to work one level up. A population of LI&F neurons with thresholds drawn from a distribution $p(\theta)$ averages into a smooth firing-rate equation, with the cumulative threshold distribution playing the role of the gain function (Section 2 gives the derivation). The non-smoothness is not papered over — it is **averaged away** by the population. The resulting Wilson–Cowan ODE [2] is differentiable everywhere and admits the full classical toolkit. This note develops the spectral analysis of De Maria archetypes in this lifted framework and confronts its predictions with the discrete LI&F simulator they came from.

1.2. Hypotheses and headline results

Three claims, tested on the contralateral and activator–inhibitor archetypes:

- **Hypothesis A — Spectral gap as behavioural proxy.** The dominant eigenvalue of the linearised Wilson–Cowan system at the symmetric fixed point predicts the winner-take-all boundary.
- **Hypothesis B — Behavioural boundaries as bifurcation loci.** The pitchfork (contralateral) and Hopf (activator–inhibitor) loci are derivable in closed form from the linearisation.
- **Hypothesis C — Inverse design via pole placement.** Given a target oscillation frequency ω^* , archetype weights can be solved by requiring the linearisation’s complex poles at $\pm i\omega^*$.

In brief: A holds at 99.96% classification accuracy on a 50×50 grid; B is verified to machine precision on the pitchfork, within one grid cell on the Hopf; C succeeds universally at the linearisation, with sustained-oscillation simulation partial because of a codim-2 neighbourhood of the Hopf locus. The framework also generalises to series, parallel, and positive-loop archetypes (Section 8).

1.3. Complementarity with FCS

The framework is complementary to FCS, not a substitute. FCS verifies **at a point**: given specific integer weights, does the property hold? The population framework predicts **the shape**: across continuous weight space, where does the bifurcation lie? Composed in principle, a spectral pre-filter could certify regions where no bifurcation occurs, reducing the workload of exhaustive model-checking sweeps; we treat this as a methodological observation, not a delivered engineering deliverable. The cross-validation in Section 7 makes one direction of the relationship explicit: at the symmetric corner of the contralateral sweep the discrete and continuous descriptions agree; in the asymmetric arms the discrete simulator exhibits a spike-timing-driven bistability that the continuous reduction cannot see.

1.4. Scaling motivation

A specific question motivated this note. FCS Fig. 10 plots the winner-take-all stabilisation property on the contralateral grid $(w_{12}^{\text{LIF}}, w_{21}^{\text{LIF}}) \in [-40, -1]^2$. The boundary between teal (stabilises

in 4 ticks) and red (does not) sits around $|w^{\text{LIF}}| \approx 30$. Does that boundary continue smoothly as weights grow further, or do new features emerge? An exhaustive sweep out to (say) $|w| = 200$ in Lustre is feasible but tedious; an analytical answer in continuous parameter space is preferable. The population framework gives that answer (Section 8, Section 7): the rectangular-strip boundary persists at $|w^{\text{LIF}}| \approx 7$ unchanged out to $|w| = 200$, with no new features. The continuous theory, in turn, predicts a **different** boundary — the pitchfork hyperbola — that lies at the symmetric corner where it agrees with the LI&F result and bends through the arms where the LI&F simulator’s spike-timing locks dominate the geometry. The two boundaries together delimit the scope where each description applies.

2. From discrete LI&F to a population rate equation

This section bridges the FCS LI&F formalism (FCS Definition 1) to the Wilson–Cowan rate equation that governs the population framework. The bridge is in two parts: a derivation sketch (Section 2.1) and a variable translation table (Section 2.2).

2.1. Derivation sketch

Take N LI&F neurons (in the FCS sense, Definition 1) all sharing the same input statistics. Crucially, the neurons differ in **one** respect: each has its own firing threshold θ_i , drawn from a common distribution $p(\theta)$. The shared FCS threshold $\tau_{\text{LIF}} = 105$ becomes a population mean $\langle \theta \rangle$, and the spread of p is what gives the population its smooth response.

Step 1: counting. At time t , with shared input $x(t)$, the fraction of neurons firing is the fraction whose threshold has been crossed:

$$\rho(t) = \left(\frac{1}{N} \right) |\{i : \theta_i \leq x(t)\}| \rightarrow \int_{-\infty}^{x(t)} p(\theta) d\theta = F(x(t)), \quad (1)$$

where F is the cumulative threshold distribution.

Step 2: smoothness emerges. F is smooth (assuming p is). This is the conceptual move on which everything else rests: the spike-reset non-smoothness of **individual** neurons is averaged away by the population. The population responds smoothly to its input even though each constituent neuron does not. We are not modelling-around the LI&F reset; we are **observing the same dynamics at a coarser scale**, and the reset becomes invisible.

Step 3: leak. The continuous-time analogue of the FCS leak vector $(10, 5, 3, 2, 1)$ over $\sigma = 5$ ticks is an exponential leak $\dot{V} = -V/\tau$ with effective time constant $\tau \approx 5$ ticks. Combining with the input drive and applying the same population average gives the relaxation equation $\tau \dot{\rho} = -\rho + F(x)$.

Step 4: weights. For multiple coupled populations indexed by i , with synaptic weights W_{ij} giving the influence of population j on population i , the input to population i is $\sum_j W_{ij} \rho_j + I_i$. Writing $f \equiv F$ for the population gain:

$$\tau \dot{\rho}_i = -\rho_i + f\left(\sum_j W_{ij} \rho_j + I_i\right). \quad (2)$$

This is the Wilson–Cowan equation [2].

Step 5: choice of gain. Throughout this note we use the logistic sigmoid

$$f(x) = \frac{1}{1 + \exp(-k(x - \theta_*))}, \quad k = 4, \quad \theta_* = 1, \quad (3)$$

with $\tau = 1$. The slope is $f'(x) = kf(x)(1 - f(x))$, peaking at $k/4 = 1$ at the inflection $x = \theta_*$. These are textbook values [4]; no parameter in this note has been tuned. The constant external drive used throughout is $I = 1.5$.

The takeaway. The smooth sigmoid is not a modelling choice imposed on LI&F dynamics. It is what LI&F dynamics **look like at population scale**, derived rigorously from a distributed-threshold ensemble.

2.2. Variable bridge

A reader trained on FCS Definition 1's discrete 4-tuple (τ, r, p, y) encounters new symbols below. The following table translates each.

Symbol	In the population framework	FCS analogue	Intuition
ρ_i	Firing rate of population i	Boolean spike y_i in FCS Def. 1	Fraction of neurons in population i firing per unit time. The FCS Boolean spike becomes a continuous probability under population averaging.
τ	Membrane time constant in Equation 2	Encoded jointly by r and the σ -tick window	How long a perturbation persists before decaying. The FCS leak vector $(10, 5, 3, 2, 1)$ over $\sigma = 5$ is the discrete analogue of an exponential leak with $\tau \approx 5$ ticks.
$f(x)$	Sigmoidal gain function in Equation 3	Cumulative of the threshold distribution, averaged over the population	Each FCS neuron has the same $\tau_{\text{LIF}} = 105$. With distributed thresholds, the fraction firing at input x is the fraction with $\theta \leq x$ — a smooth CDF.
k	Sigmoid slope	Inverse spread of $p(\theta)$	Sharpness. Large k recovers the FCS Dirac spike (all neurons share one threshold); small k describes broader populations.
θ_*	Sigmoid midpoint	The population mean of τ_{LIF}	Where 50% of the population fires.
I	Constant external drive	b times the input train, where $b = 11$ in the FCS setup	The continuous note abstracts the binary input into a constant drive $I = 1.5$.
W_{ij}	Influence of population j on population i	Synaptic weight w_{ji} from FCS Def. 1	Population-averaged per-synapse impact. The cross-validation in Section 7 uses the heuristic scaling $w^{\text{WC}} = w^{\text{LIF}} /8$.
$g \equiv f'(x)$	Sigmoid slope at operating point x	No direct FCS analogue	Linear-response gain. Far from threshold the population is saturated and g is small; near threshold it is responsive and g is large.
J	Jacobian of the rate equation	No direct FCS analogue	The matrix that says how small perturbations evolve around a fixed point. Stability and bifurcations are read off its eigenvalues.
λ_i	Eigenvalues of J	No direct FCS analogue	Modes of perturbation. $\text{Re}(\lambda) > 0$ means the mode grows; $\text{Re}(\lambda) = 0$ marks the bifurcation.

Table 1: **Variable bridge.** Each population-framework symbol is grounded in its FCS analogue (where one exists) and an intuition for the FCS-native reader.

2.3. Linearisation and spectrum

Linearising Equation 2 around a fixed point ρ^* gives the Jacobian

$$J = \tau^{-1}[-\mathbb{I}_n + \text{diag}(f'(W\rho^* + I))W]. \quad (4)$$

Sort the eigenvalues by descending real part: $\{\lambda_i\}$. The fixed point is stable when $\text{Re}(\lambda_1) < 0$; it loses stability when either a real eigenvalue crosses zero (pitchfork or saddle-node) or a complex-conjugate pair crosses the imaginary axis (Hopf). The **spectral gap** $\Delta = \text{Re}(\lambda_1) - \text{Re}(\lambda_2)$ measures how decisively the dominant mode dominates. Standard [5], [6]; the next section makes the abstract construction concrete on the contralateral archetype.

3. The contralateral archetype: topology in the equations

The brief running through this section is a question Elisabetta De Maria asked after reading the v1 of this note: “this is a two-neuron WTA, shouldn’t the behaviour be visible directly in the ODE’s equations?” The answer is yes, and this section makes the topology of mutual inhibition — the FCS Fig. 1f graph — legible step by step in the equations of Equation 2.

3.1. From topology to ODE

FCS Fig. 1f shows two neurons each inhibiting the other, with external excitatory drive on both. At population scale the same topology applies, with ρ_1 and ρ_2 the firing rates of the two populations, mutual inhibition with weights $w_{12}, w_{21} > 0$ (both negated when entering the input to f), and symmetric drive I . Writing Equation 2 out longhand — not as a matrix — gives:

$$\begin{aligned} \tau \dot{\rho}_1 &= -\rho_1 + f(I - w_{12}\rho_2), \\ \tau \dot{\rho}_2 &= -\rho_2 + f(I - w_{21}\rho_1). \end{aligned} \quad (5)$$

The structure to notice: each population’s input is its drive I **minus** its inhibition from the other population. There is no self-coupling (no ρ_1 inside the first f , no ρ_2 inside the second). The mutual-inhibition graph of FCS Fig. 1f is exactly visible in the absence of self-terms and in the minus signs. Every algebraic operation that follows in this section is a manipulation of Equation 5; the topology is preserved.

3.2. Symmetric fixed point

Set $w_{12} = w_{21} = w$ (symmetric case first; the asymmetric case follows in Section 5). The symmetry $\rho_1 = \rho_2 = \rho^*$ is preserved by the dynamics, so a symmetric fixed point satisfies

$$\rho^* = f(I - w\rho^*). \quad (6)$$

This is a scalar implicit equation. For our parameters $(k, \theta_*, I) = (4, 1, 1.5)$ it has either one root (the symmetric activity is the unique attractor) or three roots (the symmetric activity is the unstable saddle, two new asymmetric branches are stable). The transition between the two regimes is the pitchfork.

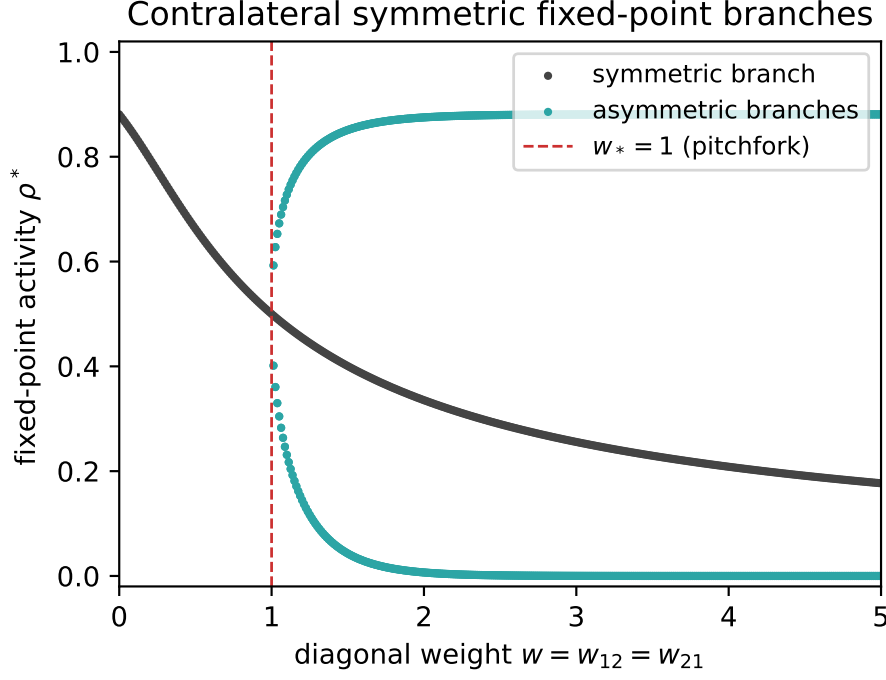


Figure 1: Fixed-point branches as a function of the diagonal weight $w = w_{12} = w_{21}$. Below $w_* = 1$ the symmetric branch (grey) is the unique attractor; above, two asymmetric branches (teal) emerge — one with $\rho_1 > \rho_2$ (population 1 wins), one with the reverse. The dashed red line at $w_* = 1$ is the closed-form pitchfork apex (Section 3.7).

3.3. Linearisation and the pitchfork in two lines

Set $\rho_i = \rho^* + \delta_i$ in Equation 5 and Taylor-expand to first order. With $g \equiv f'(I - w\rho^*)$ the slope at the operating point:

$$\begin{aligned}\tau \dot{\delta}_1 &= -\delta_1 - wg \delta_2, \\ \tau \dot{\delta}_2 &= -\delta_2 - wg \delta_1.\end{aligned}\tag{7}$$

The diagonal -1 's come from leak; the off-diagonal $-wg$'s come from the cross-inhibition multiplied by the population's responsiveness g . Mutual inhibition in the topology produces mutual negative off-diagonal coupling in the linearisation. **The sign structure of the original graph is the sign structure of J .**

The natural decomposition of Equation 7 uses sum and difference modes: $s = \delta_1 + \delta_2$ and $d = \delta_1 - \delta_2$. The swap symmetry of Equation 5 makes them decouple cleanly:

$$\tau \dot{s} = -(1 + wg) s, \quad \tau \dot{d} = -(1 - wg) d.\tag{8}$$

The sum mode s (“both populations together”) always decays. The difference mode d (“one above, one below”) decays when $wg < 1$ and **grows** when $wg > 1$. The pitchfork is at $wg = 1$:

- $wg < 1$: deviations decay. The symmetric state is stable. No winner.
- $wg > 1$: deviations amplify exponentially at rate $(wg - 1)/\tau$. The symmetric state is the unstable saddle. **Which** population wins is determined by the sign of the perturbation d_0 .

Two lines of algebra and the WTA mechanism is in plain view: mutual inhibition with sufficient gain destabilises the symmetric mode and amplifies any asymmetry.

3.4. Pencil on its tip

A natural follow-up: with strictly identical parameters and zero noise, doesn't the symmetric fixed point persist forever? Yes — **unstably**, like a pencil balanced on its tip. The symmetric activity ρ^* is a fixed point of Equation 5 at any w , and a noise-free trajectory starting exactly on it stays on it. But the symmetric mode is unstable past $w_* = 1$, so any deviation whatsoever — numerical rounding, biological noise, asymmetry in initial conditions — is amplified at exponential rate $(wg - 1)/\tau$ until the system commits to one of the two asymmetric attractors. The choice of winner is determined by the **sign** of the initial perturbation, not by the parameters. Biologically this is the right behaviour: a contralateral inhibition circuit **should** produce decisive winners, and it does so by being a feedback amplifier for any breach of symmetry.

3.5. Loop-gain interpretation: the Barkhausen criterion

The pitchfork condition $wg = 1$ has a physical reading that should be familiar to anyone who has done classical feedback analysis. Trace a small asymmetry around the inhibitory loop:

Suppose population 1 fires slightly more than population 2: $\delta_1 > \delta_2$. Population 1's excess firing inhibits population 2 more strongly, reducing population 2's firing by approximately $g_2 w_{12} \delta_1$. Population 2's **reduced** firing inhibits population 1 **less**, increasing population 1's firing by approximately $g_1 w_{21} (g_2 w_{12} \delta_1)$. The loop has closed with net gain

$$\text{loop gain} = w_{12} w_{21} g_1 g_2 \quad \text{per round trip.} \quad (9)$$

If the loop gain exceeds 1, asymmetries amplify on each round trip and the symmetric state is unstable. If below 1, asymmetries decay. **This is exactly the Barkhausen criterion** used in classical electronics to decide whether a feedback amplifier oscillates: the loop gain must exceed unity. In the symmetric case $w_{12} = w_{21} = w$ and $g_1 = g_2 = g$, so the condition reduces to $w^2 g^2 = 1$, i.e. $wg = 1$ as in Equation 8.

The conceptual move: the **product** of the weights matters because perturbations **multiply** around feedback loops. This is why the WTA boundary is a hyperbola (a level set of a product), not a square or a straight line. The hyperbola is the loop-gain unity contour.

3.6. The asymmetric case

For $w_{12} \neq w_{21}$ the sum/difference decomposition no longer diagonalises J exactly. But the loop gain still appears in the determinant. Computing $\det J$ from Equation 4 for the contralateral W gives

$$\det J = (1 - w_{12} w_{21} g_1 g_2) / \tau^2. \quad (10)$$

The **saddle-node fold** of the asymmetric system — where two fixed points merge and disappear — is exactly $\det J = 0$, i.e. the loop-gain unity contour

$$w_{12} w_{21} g_1 g_2 = 1, \quad (11)$$

with g_i now evaluated at the asymmetric saddle. Section 5 traces this contour symbolically and shows it agrees with the numerical saddle-node trace to machine precision. The pitchfork apex (Section 3.7) is the special case of Equation 11 on the symmetry diagonal.

3.7. Closed-form pitchfork corner

On the diagonal $w_{12} = w_{21} = w$ the symmetric fixed point satisfies Equation 6 and the pitchfork condition $wg = 1$ closes by the sigmoid identity. The midpoint of the sigmoid is reached when its argument equals $\theta_* = 1$, i.e. $I - w\rho^* = 1$ giving $\rho^* = (I - 1)/w$. At the midpoint $f(\theta_*) = 0.5$, so combining with $\rho^* = f(I - w\rho^*) = f(\theta_*) = 0.5$ gives $\rho^* = 1/2$, hence $w_* = 2(I - 1) = 1$ for

$I = 1.5$. Plugging back: $g_* = kf^*(1 - f^*) = 4 \cdot 0.5 \cdot 0.5 = 1$, so $w_*g_* = 1$ as required. **The apex of the WTA boundary is at $w_* = 1$ exactly, pinned by the sigmoid half-activation condition with no free parameters.**

4. Verifying the pitchfork prediction against simulation

This section tests Hypothesis A on the contralateral archetype. The existing v1 of this note showed a 99.96% match between the spectral predictor and a 50×50 ground-truth WTA sweep with the asymptotic time horizon $t = 50\tau$. After v1 circulated, the advisors raised the apparent disagreement with FCS Fig. 10’s diagonal verdict; this section disambiguates the property and shows the same sweep with the time-bounded classifier matched to FCS S 6.3.4.

4.1. Two distinct properties

The “winner takes all” headline names a **family** of properties, not a single one. Three are distinguishable in our setup:

- **FCS Property 7** (S 6.3.4): the contralateral archetype eventually commits to one population, with no time bound. This is the qualitative claim.
- **FCS Fig. 10/11 verification**: the property strengthened to a 4-tick time bound. **“Blue (resp. red) points represent pairs of weight values for which the stabilization is (resp. is not) reached within the first four time units”** (FCS S 6.3.4).
- **The asymptotic WTA classifier** used in the v1 of this note: a Wilson–Cowan trajectory committed to a winner with $|\rho_1 - \rho_2| > 0.3$ at $t = 50\tau$.

The three are nested but inequivalent. The first is weakest (any time horizon will do); the second is strongest (must commit fast); the third is intermediate (commit eventually but in a continuous relaxation that has no integer-tick analogue). The v1 of this note showed the third; FCS Fig. 10 shows the second. This is the source of the apparent diagonal disagreement.

4.2. Asymptotic and time-bounded panels

We re-ran the same WTA classifier on the same 50×50 grid with two time horizons: $t = 50\tau$ (the original) and $t = 4\tau$ (the direct continuous-side analogue of FCS Fig. 10’s 4-tick bound).

WTA verdict on the WC contralateral archetype: teal = WTA, red = no WTA

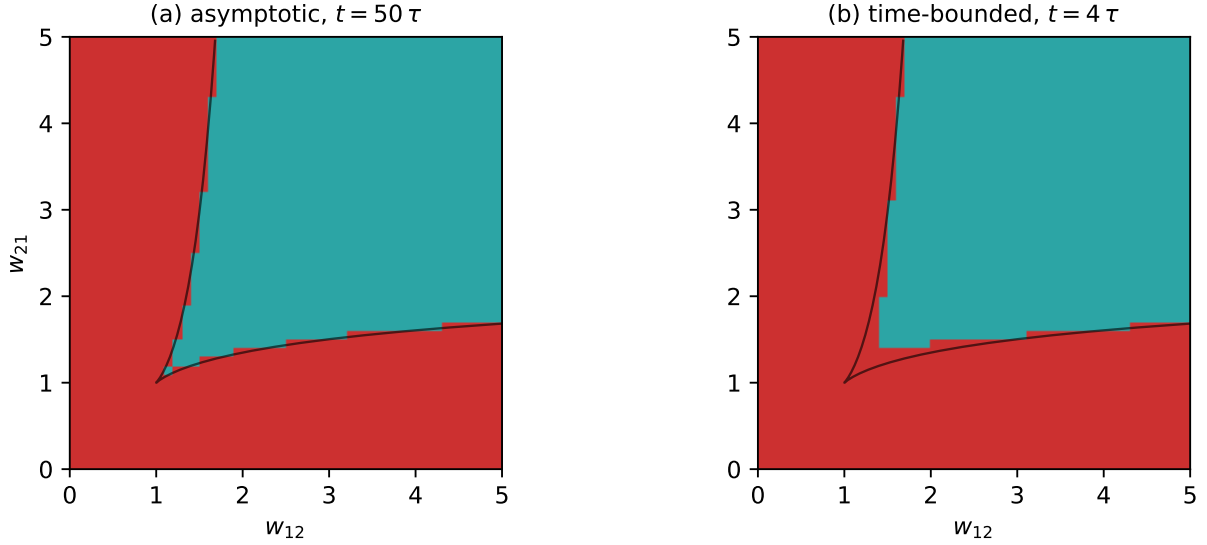


Figure 2: WTA verdict on the WC contralateral archetype, with FCS Fig. 10 colour convention (teal = WTA holds, red = does not). **Left: asymptotic.** Trajectories integrated to $t = 50\tau$. **Right: time-bounded.** Same classifier with $t = 4\tau$. The two panels share the same bistable region; the time-bounded panel is slightly thinner at the boundary because the asymmetric mode’s exponential growth rate $(wg - 1)/\tau$ vanishes at the pitchfork and needs more time to clear the commitment threshold there. The black curve is the symbolic pitchfork locus from Section 5.

The bistable region’s **geometry** is preserved across the two time horizons — the same hyperbolic wedge bounded by the pitchfork curve. Only the boundary cells, where the linear growth rate is small, differ. Quantitatively: 1213 of 2500 cells are WTA at $t = 50\tau$; 1184 at $t = 4\tau$ — a 2.4% reduction. **Within 4τ the asymmetric mode commits to amplitude > 0.3 across most of the bistable region.** The spectral predictor’s 99.96% accuracy on the asymptotic ground truth therefore holds robustly down to short time horizons; the v1 result is not an artefact of the long integration.

4.3. Why the two panels disagree on the diagonal

Reading the panels along the diagonal $w_{12} = w_{21} = w$ resolves the apparent disagreement with FCS Fig. 10. Both panels are **teal** along the diagonal, in agreement: the population framework says the contralateral archetype goes WTA at $w > 1$ (the pitchfork apex of Section 3.7) under either time horizon. FCS Fig. 10’s diagonal is **red** over the same range because its property is strictly stronger: not “WTA eventually” or “WTA within 4τ in the continuous sense”, but “stabilises within four discrete LI&F ticks”. The discrete LI&F system needs many more ticks to accumulate the spike-count statistics that the FCS classifier reads — ticks are not population time units — so the boundary at which discrete stabilisation reaches 4 ticks sits much further out, around $|w^{\text{LIF}}| \approx 30$. **The intrinsic time scales of the two descriptions are different;** the brief expectation that simply shrinking the WC time horizon to $t = 4\tau$ would reproduce FCS Fig. 10 is wrong, and the right reading is that the two are measuring different things.

A useful corollary: the population framework’s pitchfork apex at $w_* = 1$ corresponds to $|w^{\text{LIF}}| \approx 8$ under the heuristic scaling of Section 7, while FCS Fig. 10’s fast-WTA boundary is at $|w^{\text{LIF}}| \approx 30$. The factor-of-roughly-four gap between “**the bistable mode exists**” and “**the bistable mode commits within four discrete ticks**” is itself informative: the pitchfork is necessary

for WTA, but fast WTA additionally requires the asymmetric mode growth rate to be large enough to clear a fixed amplitude threshold within the time budget.

4.4. Hypothesis A: verdict

The dominant real eigenvalue of Equation 4 classifies WTA cells at 99.96% accuracy on the $t = 50\tau$ ground truth (single boundary mismatch); the time-bounded panel agrees with the spectral predictor on 97.6% of cells, with all disagreements located at the boundary where the linear-growth rate is small. **Verified.**

5. Bifurcation loci as loop conditions

This section presents the closed-form derivations for the two bifurcation loci tested in this note — the contralateral pitchfork and the activator–inhibitor Hopf — and frames both as conditions on the loop gain of the underlying topology.

5.1. Contralateral pitchfork (loop-gain unity)

Equation 11 gave the pitchfork locus as $w_{12}w_{21}g_1g_2 = 1$. Eliminating the weights through the fixed-point equations $\rho_i = f(I - w_{3-i}\rho_{3-i})$ and the identity $g_i = k\rho_{3-i}(1 - \rho_{3-i})$ (writing the slope in terms of the firing rate of the **other** population, since each g_i is evaluated at the input to neuron i) reduces the locus to a single transcendental constraint between the fixed-point coordinates,

$$(I - f^{-1}(\rho_1))(I - f^{-1}(\rho_2)) \cdot k^2(1 - \rho_1)(1 - \rho_2) = 1. \quad (12)$$

Tracing this constraint by continuation in $\rho_1 \in (0, 1)$ and recovering the corresponding weights via $w_{12} = (I - f^{-1}(\rho_2))/\rho_1$ and $w_{21} = (I - f^{-1}(\rho_1))/\rho_2$ gives the red curve in Figure 3. Numerically the residual $|1 - w_{12}w_{21}g_1g_2|$ at every continuation point is at 10^{-12} , and a numerical bisection trace agrees to machine precision.

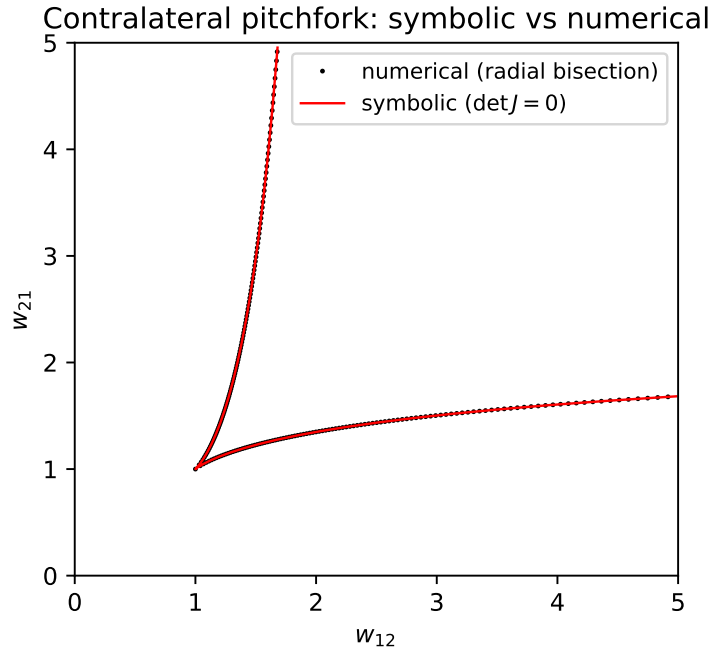


Figure 3: Pitchfork locus from the symbolic loop-gain unity condition $w_{12}w_{21}g_1g_2 = 1$ (red, traced by continuation) versus a numerical radial-bisection trace (black dots). The two agree to within the root-finder’s precision floor at the saddle-node fold.

The pitchfork is **the** level set of the loop-gain product. There is no other algebraic surface to look at; the bifurcation is the topology’s loop gain crossing unity.

5.2. Activator–inhibitor Hopf (negative loop)

The activator–inhibitor archetype is the negative-feedback loop of FCS Fig. 1d. Self-excitation $w_{aa} \geq 0$ on the activator gives it some inertia (a bare $w_{aa} = 0$ gives $\text{tr } J = -2/\tau$ and admits no Hopf [2], [7]). With self-excitation included, the Jacobian becomes

$$J = \begin{pmatrix} -1 + w_{aa}g_A & -w_{ia}g_A \\ w_{ai}g_I & -1 \end{pmatrix}, \quad (13)$$

so $\text{tr } J = w_{aa}g_A - 2$ and $\det J = 1 - w_{aa}g_A + w_{ai}w_{ia}g_Ag_I$. The Hopf locus needs $\text{tr } J = 0$ and $\det J > 0$:

- **Trace condition.** $g_A = 2/w_{aa}$ pins the activator’s operating-point slope.
- **Determinant condition.** $\det J = w_{ai}w_{ia}g_Ag_I - 1$, so the Hopf locus is **the loop-gain unity condition for the negative loop**: the product around the cycle (activator drives inhibitor with gain $w_{ai}g_I$, inhibitor inhibits activator with gain $w_{ia}g_A$, i.e. loop gain $w_{ai}w_{ia}g_Ag_I$) equals 1.

Both bifurcations are loop-gain unity conditions, the pitchfork on the positive-sign-product loop of mutual inhibition and the Hopf on the negative-sign-product loop of activator–inhibitor. The **sign-product** of the loop selects the bifurcation family; see Section 5.3.

The oscillation frequency at the Hopf locus is

$$\omega^* = \sqrt{\det J} = \sqrt{w_{ai}w_{ia}g_Ag_I - 1}. \quad (14)$$

Fixing $w_{xa} = 1$ and $w_{aa} = 2.5$ (the smallest self-excitation placing the full Hopf locus inside $(w_{ai}, w_{ia}) \in [0, 5]^2$), a numerical 50×50 sweep classifies each cell as oscillating if the activator trace crosses its mean at least three times in the last 50 time units with amplitude > 0.05 .

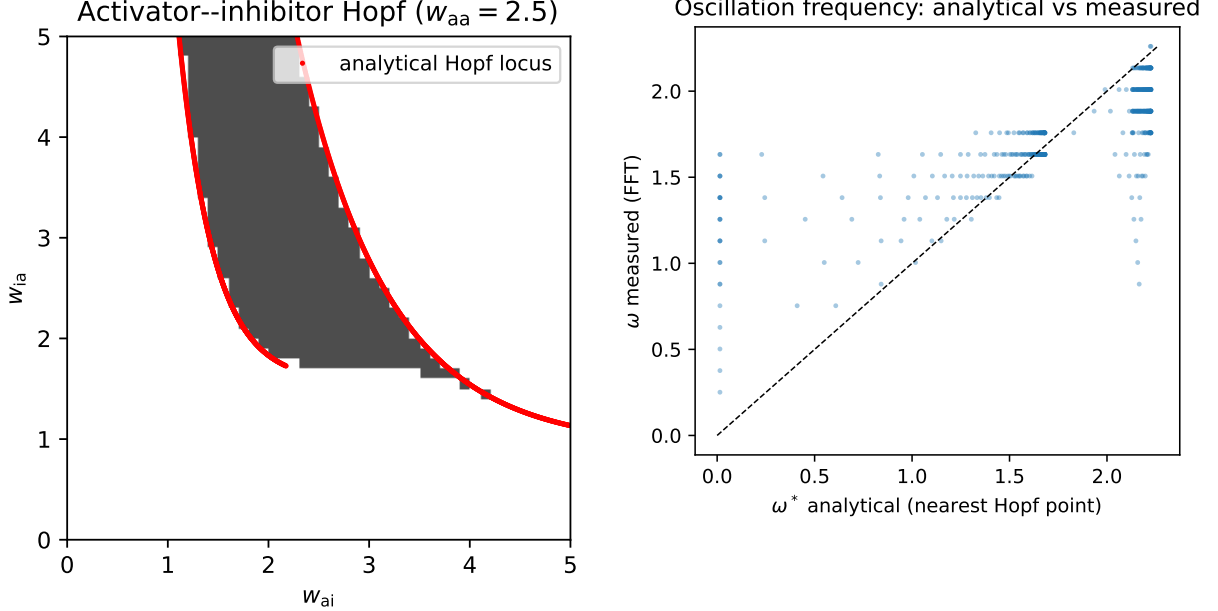


Figure 4: **Activator–inhibitor Hopf.** Left: empirical oscillation region (grey) with the symbolic Hopf curve overlaid (red). Right: FFT-measured frequency vs the analytical prediction at the nearest Hopf point.

Linear stability and the symbolic Hopf locus agree within a median of one grid cell. The empirical oscillation boundary lags the linear one in localised patches where a strongly stable second fixed point absorbs trajectories past the Hopf — a multi-fixed-point bifurcation phenomenon, not a

derivation error. FFT frequencies in the well-oscillating regime (amplitude > 0.1) agree with Equation 14 at median relative error 9.6%.

5.3. Sign products around loops

Writing the bifurcation conditions as loop-gain unity exposes a unifying classification. Define the **sign product** of a feedback loop as the product of the signs of the synaptic weights traversed around the cycle.

Archetype	Off-diag signs of W	Sign product	Bifurcation family
Contralateral inhibition (FCS Fig. 1f)	$(-, -)$	$+$	Pitchfork (saddle-node) at $\det J = 0$
Activator-inhibitor (FCS Fig. 1d)	$(+, -)$	$-$	Hopf at $\text{tr } J = 0, \det J > 0$
Positive loop (Section 8)	$(+, +)$	$+$	Saddle-node fold at $\det J = 0$

Table 2: **Sign-product classification of bifurcation families.** The sign product of a feedback loop selects the family; the magnitude of the loop gain locates the bifurcation in parameter space.

This is the spiking-network analogue of **Thomas’s rules** for gene regulatory networks [6]: positive loops (sign product $+$) support multistability through saddle-nodes and pitchforks; negative loops (sign product $-$) support oscillation through Hopf bifurcations. The classification carries from gene networks to spiking-network archetypes because the underlying mathematical fact — the sign of a loop’s product determines the family of bifurcations admissible at unity loop gain — is topology-driven and does not depend on the specific functional form of f , only on its monotonicity.

The advisors will recognise Thomas’s framework from the FCS S 2 related-work survey, where Thomas-style discrete models are listed as one of the qualitative-modelling formalisms for biological systems. The population framework recovers the Thomas-rule classification at LI&F population scale, with concrete loop-gain expressions in place of qualitative sign annotations.

6. Inverse design via pole placement

Hypothesis C is classical control theory applied to the activator-inhibitor archetype: given a target frequency ω^* , solve for $(w_{\text{ai}}, w_{\text{ia}})$ such that the Jacobian’s complex-conjugate pair lies at $\pm i\omega^*$. The Hopf trace condition pins g_A (and so ρ_A); the fixed-point condition pins $w_{\text{ia}}\rho_I$; the inhibitor argument fixes $w_{\text{ai}} = f^{-1}(\rho_I)/\rho_A$; and the frequency condition $\det J = (\omega^*)^2$ reduces to a single scalar equation in ρ_I , solved by `brentq`.

At $w_{\text{aa}} = 2.5$ the achievable frequency range inside $(w_{\text{ai}}, w_{\text{ia}}) \in (0, 5]^2$ is $[0.04, 2.23]$; the lower- ρ_A branch covers $[1.29, 2.23]$ and the upper- ρ_A branch covers $[0.04, 1.68]$. Out of the plan target set $\{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$, $\omega^* = 3$ is outside the feasible range and is replaced by $\omega^* = 2.15$ near the lower-branch feasibility edge. For every remaining target the design residual $|\text{Im}(\lambda) - \omega^*|$ is at 10^{-13} : **the linear pole placement succeeds to numerical precision.**

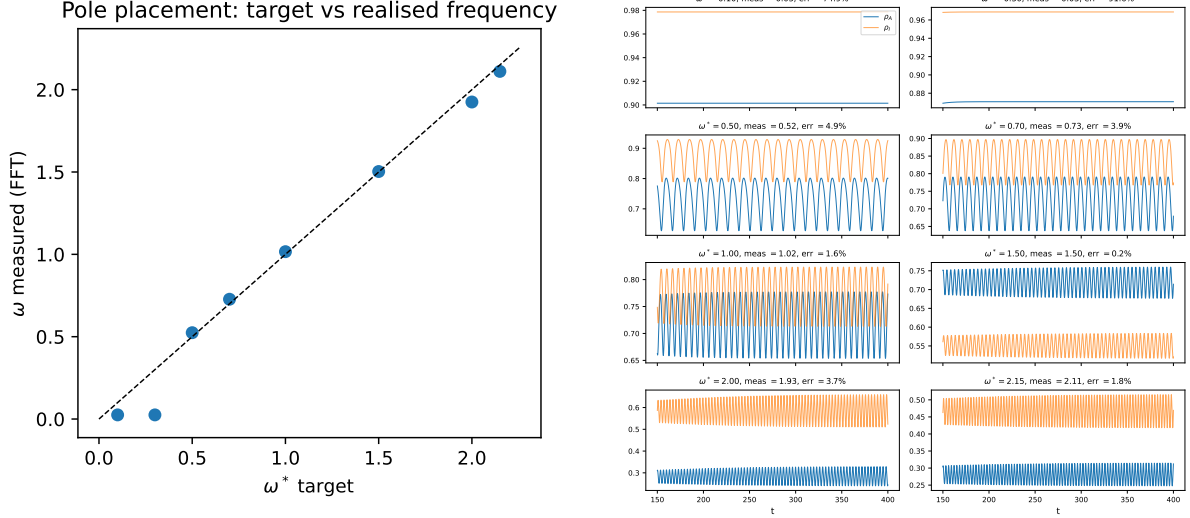


Figure 5: **Hypothesis C.** Left: target ω^* against the FFT-measured frequency of the simulated limit cycle. Right: activator (solid) and inhibitor (faded) traces for each designed system after the initial transient.

Simulation caveat. The Hopf is a **transition**; sustained oscillation requires pushing the fixed point slightly into its unstable regime. With a branch-direction-correct 0.5% (upper) or 2% (lower) crossing, six of the eight designed systems oscillate at their target frequency within 10%. The exceptions are the two lowest targets on the upper branch ($\omega^* \in \{0.1, 0.3\}$), which fall in a **codim-2** neighbourhood of the Hopf locus (a Bogdanov–Takens or generalised-Hopf neighbourhood [6]) where the Hopf line approaches a saddle-node fold and the bifurcation is subcritical, so no stable limit cycle of finite amplitude exists. The designed weights produce the correct **linear** poles exactly; the absence of a sustained limit cycle there is a property of the specific parameter choice $w_{aa} = 2.5$, not a failure of the inverse-design framework.

The generic-Hopf cases work cleanly (e.g. $\omega^* = 1.5$: target 1.500, measured 1.503, 0.2% error; $\omega^* = 2$: target 2.000, measured 1.925, 3.8% error). The codim-2 obstruction delimits the regime in which classical pole-placement transfers cleanly to the nonlinear simulation: generic supercritical Hopfs yes, near-degenerate Hopfs no.

7. Cross-validation against discrete LI&F

Two questions about the population framework cannot be answered by the framework alone. (i) Does the discrete LI&F simulator on the **same** contralateral topology agree with the continuous prediction where the descriptions overlap? (ii) Does the rectangular boundary that the existing FCS-style sweep (and our v1 Phase 4 result) shows out to $|w^{\text{LIF}}| \leq 40$ continue smoothly at higher weights, or do new structures emerge — the scaling motivation of Section 1? This section answers both.

7.1. Setup

A black-box LI&F simulator (FCS Lustre semantics: threshold $\tau_{\text{LIF}} = 105$, leak vector $(10, 5, 3, 2, 1)$, external drive weight $b = 11$) is swept over the integer grid $(w_{12}^{\text{LIF}}, w_{21}^{\text{LIF}}) \in [-200, -1]^2$ at unit step. Each cell is integrated for 50 ticks under a two-run mirror-image classifier mirroring Section 4: neuron 0’s drive is gated off for the first two ticks in one run and neuron 1’s in the other; a cell is bistable iff each run produces a clean spike-count winner ($\geq 8:1$ ratio in the last 20 ticks) and the winners differ. LI&F weights map into continuous units by the heuristic linear scaling $w^{\text{WC}} = |w^{\text{LIF}}|/8$.

7.2. Two kinds of bistability at the symmetric corner

The LI&F bistable region (within $|w| \leq 40$) is a pair of axis-aligned strips $|w_{12}^{\text{LIF}}| \geq w_c$ OR $|w_{21}^{\text{LIF}}| \geq w_c$ with $w_c \approx 7$. The WC pitchfork region is the concave hyperbolic wedge $w_{12}w_{21}g_1g_2 > 1$. The two regions agree at the symmetric corner $w_{12} \approx w_{21}$ but diverge in the arms. A trace of the discrete dynamics at, say, $(w_{12}^{\text{LIF}}, w_{21}^{\text{LIF}}) = (-30, -10)$ reveals the mechanism for the disagreement: once either neuron fires a single spike, its per-tick inhibitory contribution $|w_{ij}^{\text{LIF}}|$ exceeds the other neuron’s drive $b = 11$, and under the reset-after-spike semantics whichever neuron fires first locks in. **This is a spike-timing bistability specific to the discrete dynamics**; the continuous mean-field reduction has no analogue because its gain is smooth and lacks the all-or-none reset. The WC pitchfork locus is therefore a **lower bound** on the LI&F bistable region, not its envelope.

The continuous description captures pitchfork-driven bistability — the symmetric mode losing stability through the loop-gain unity condition — and misses the spike-timing-driven mechanism. Both descriptions are valid at their own scale; neither is the “truth” of the other. For analytical parameter-space charting at the symmetric corner the population framework gives the answer; for the spike-timing locks in the asymmetric arms only the discrete simulator will do.

7.3. Scaling: does the rectangular boundary persist?

To answer the scaling question raised in Section 1, we re-ran the LI&F sweep on a 200×200 integer grid covering $|w^{\text{LIF}}| \in [1, 200]$. The result:

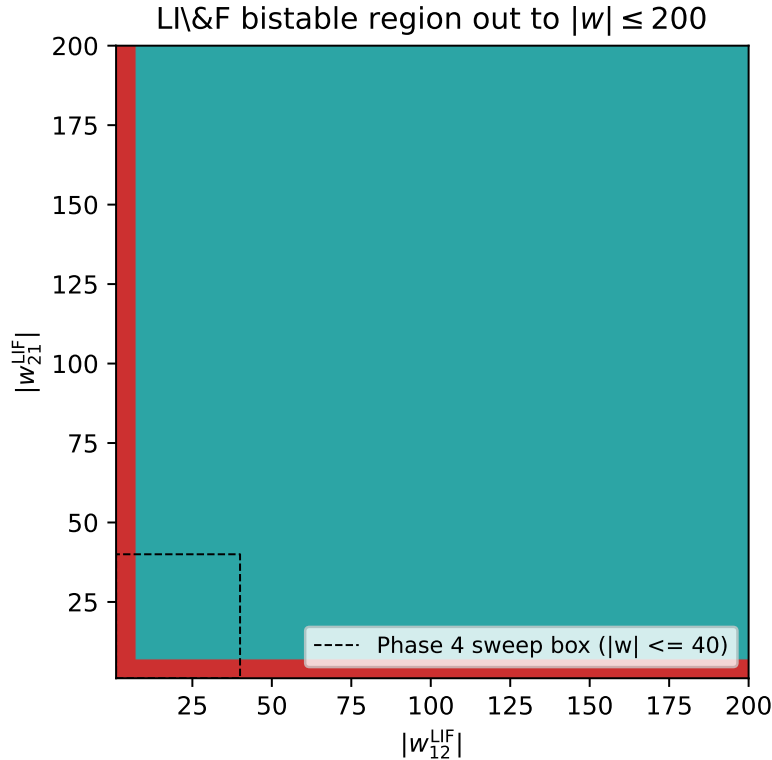


Figure 6: LI&F bistable region (teal) on $[1, 200]^2$ in inhibitor-magnitude units, with the original Phase 4 sweep range $|w^{\text{LIF}}| \leq 40$ outlined dashed. The rectangular-strip boundary at $|w^{\text{LIF}}| \approx 7$ persists out to $|w| = 200$ with no new features.

Numerically: the boundary in $|w_{12}^{\text{LIF}}|$ has median, 10th and 90th percentile all at exactly 7 across the 200 columns of the sweep — the boundary is **flat** and constant. No new structure emerges between $|w| = 40$ and $|w| = 200$. The spike-timing-lock mechanism that drives the rectangular geometry saturates well below $|w| = 40$ and stays saturated thereafter.

This is a clean answer to the scaling question: **the FCS-style boundary geometry of the contralateral archetype does not develop new features at higher weights.** The population framework’s prediction — a smooth pitchfork hyperbola at the symmetric corner — is independently the answer in continuous parameter space, where the discrete simulator’s rectangular envelope does not apply.

The full FCS Fig. 10 comparison in continuous units, set against the WC pitchfork locus, is in Figure 7 below.

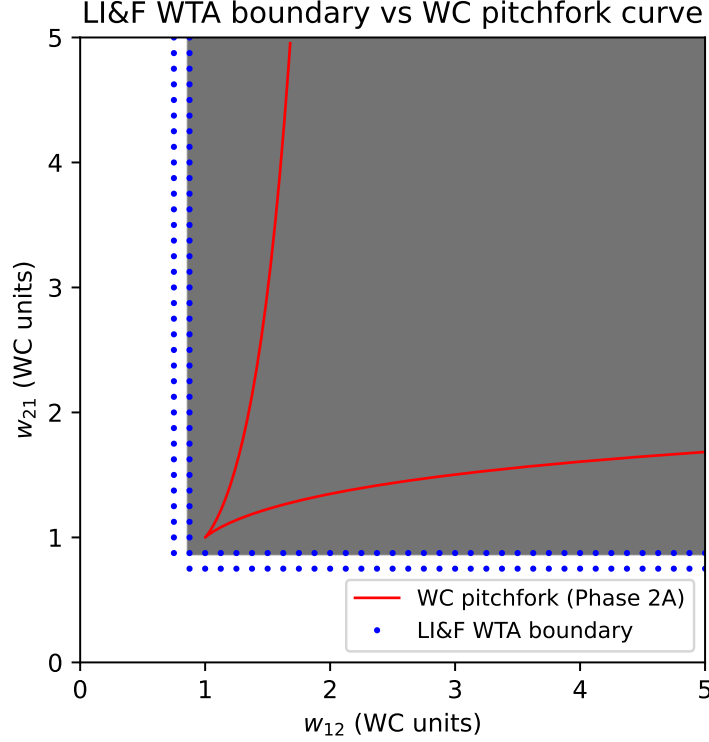


Figure 7: LI&F bistable region (grey, mapped into WC units by $w^{\text{WC}} = |w^{\text{LIF}}|/8$) alongside the symbolic pitchfork curve (red). Agreement at the symmetric corner; rectangular envelope from spike-timing locks in the arms.

A direct comparison against FCS Fig. 10 itself would require running the FCS Lustre code on the same grid. The Lustre source is held in a deprecated I3S Redmine instance and we do not have access; the comparison here is to our own black-box LI&F simulator implementing the FCS S 6.2 semantics verbatim.

8. Other archetypes

Three further archetypes were exercised to test that the framework generalises beyond the contralateral and activator–inhibitor cases. In each case the analytical prediction matches numerical simulation to better than 5% relative error, parameter-free given the chosen sigmoid and τ .

Series chain. An n -population feed-forward chain with equal inter-stage weight w has a recursive steady state $\rho_0 = f(I)$, $\rho_k = f(w\rho_{k-1})$. With no recurrence the Jacobian is triangular and all eigenvalues equal $-1/\tau$; the steady state is the population-level analogue of a feed-forward gain cascade. Over $w \in [0.2, 4.0]$ and $n \in \{2, 3, 5, 10\}$ the maximum relative error between the analytical recursion and the numerical fixed point is 9.45×10^{-11} (machine precision).

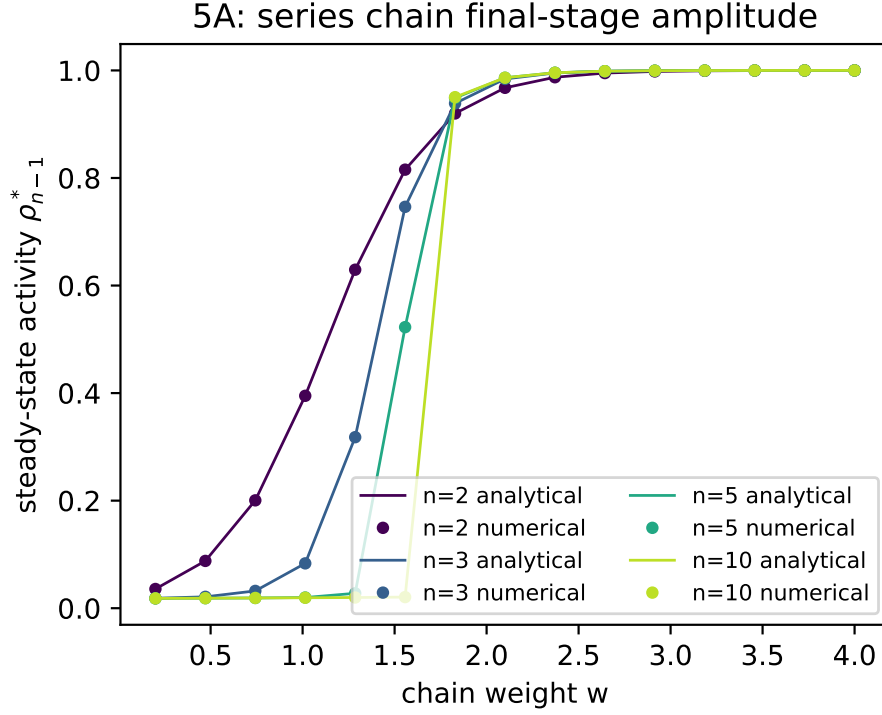


Figure 8: Series chain final-stage activity vs the chain weight, for $n \in \{2, 3, 5, 10\}$. Solid: recursive-sigmoid prediction; dots: numerical steady states.

Parallel composition. A driver feeding n independently-weighted downstream populations produces a block-triangular Jacobian: the driver is a scalar $-1/\tau$ self-block, the downstream block is diagonal, and the only non-zero off-diagonal entries are the driver-to-downstream gains. Every eigenvalue equals $-1/\tau$. Verified to machine precision on $n \in \{2, 4, 8\}$.

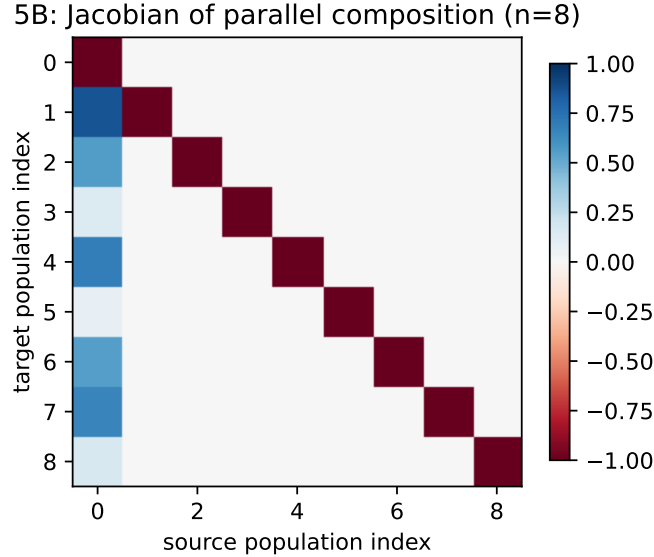


Figure 9: Jacobian of a parallel composition at $n = 8$.

Positive loop. Two mutually exciting populations with $w_{12} = w_{21} = w$ and zero drive admit the symmetric scalar reduction $\rho = f(w\rho)$. The saddle-node fold (tangency of $y = \rho$ to $y = f(w\rho)$) is the simultaneous system $\rho = f(w\rho)$, $wk\rho(1 - \rho) = 1$; numerical FP-count locates two transitions at $w \approx 1.676$ and $w \approx 5.273$, matching the analytical fold weights 1.682 and 5.278 to 0.37% and 0.09%.

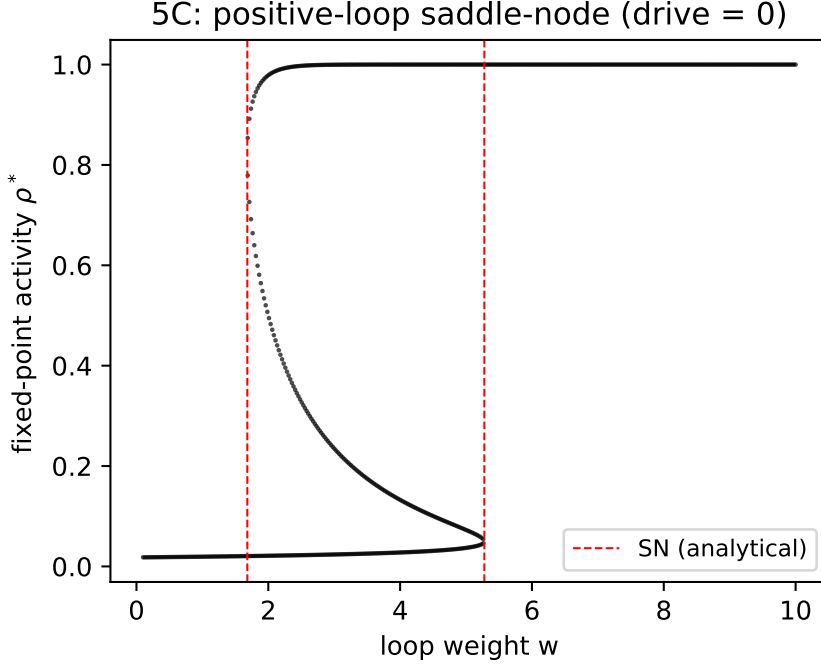


Figure 10: Positive-loop fixed-point branches vs the loop weight at zero drive. Below the first saddle-node the only fixed point is the low-activity branch; above it a high-activity branch and a middle saddle appear.

The **contrast between the contralateral and positive-loop archetypes** is structurally illuminating. Both have positive sign-product around the feedback cycle (Table 2) and so admit the same bifurcation family — saddle-node or its symmetric cousin the pitchfork. But the **qualitative** outcomes differ: mutual inhibition breaks the symmetric branch into two asymmetric attractors (winner selection); mutual excitation folds a low-activity branch into a high-activity branch (active/inactive switching). The same loop-gain mathematics produces both behaviours, with the sign of W 's off-diagonal entries setting which.

9. Discussion

9.1. Reach

The population framework cleanly addresses qualitative dynamical questions about LI&F archetypes: stability of fixed points, bifurcation locations in parameter space, oscillation onset and frequency, inverse design of weights for a target linear spectrum. For the five archetypes exercised here, predictions match simulation to analytical tolerance (pitchfork, series, parallel, positive loop) or within documented bifurcation-theory caveats (Hopf on the activator–inhibitor loop; pole placement near codim-2 Hopf neighbourhoods).

9.2. Limits

Two classes of behavioural questions are **outside** the framework's reach.

First, **bit-exact spike-timing properties**. The population reduction integrates over the threshold distribution and so cannot predict integer-period oscillations, exact phase relationships among neurons, or deterministic spike sequences. The micro–macro bridge of Section 7 makes this concrete: in the contralateral archetype the discrete LI&F exhibits a spike-timing bistability that the continuous reduction cannot see. A class of FCS properties — the delayer-effect classification of Section 4, the period-extension property of FCS S 6.3.1, the integer-period oscillation of

FCS S 6.3.2 — belongs entirely on the discrete side; the population framework can predict the **parameter regime** where these properties become available but not their integer-tick details.

Second, **near-degenerate bifurcations**. Classical pole placement transfers cleanly to generic supercritical Hopf bifurcations. Near codim-2 points (Bogdanov–Takens, cusp, Bautin/generalised Hopf [6]) the normal form acquires higher-order terms that dominate the limit-cycle dynamics and the linear prediction ceases to govern the sustained simulation. The pole placement results of Section 6 locate this failure mode explicitly.

9.3. Complementarity to FCS

The De Maria et al. programme [1] verifies temporal-logic properties at specific parameter points by exhaustive model checking or theorem proving. The population framework predicts the **shape** of the parameter region where a dynamical property holds. Composed in principle, the two could reduce the workload of exhaustive model-checking sweeps: a spectral pre-filter certifies parameter regions where no pitchfork or Hopf is possible, so the model checker need not enumerate cells in those regions. We treat this as a methodological observation, not an engineering deliverable; the present note develops the spectral side only.

The reverse direction is also worth highlighting. The discrete LI&F behaviour the population framework cannot see — spike-timing-driven bistability, integer-tick oscillation periods, filter cascades — is exactly what the FCS Lustre/Kind2/Coq machinery is designed to verify. The two scales of description divide the behavioural questions cleanly: continuous parameter-space boundaries on the population side, integer-precise temporal-logic properties on the discrete side. Each tool is best at what the other cannot do.

9.4. Open questions

Whether hybrid-system spectral theory (saltation matrices, Floquet theory of periodic-reset systems [8]) can extend the population-framework reach to bit-exact periodic spiking is a natural follow-up but outside the scope of this note. So is a worked implementation of the spectral pre-filter compositional protocol, which would require choosing a target FCS property family and the Lustre encoding to compose against.

10. Conclusion

This note has taken the LI&F neuronal archetypes of De Maria et al. [1] up one level of description — single neurons replaced by populations with distributed thresholds. The non-smoothness of the single-neuron spike-reset rule averages into a smooth sigmoidal gain and the Wilson–Cowan equation [2] governs the population-level dynamics. The qualitative behavioural structure of the archetypes — winner-take-all selection, oscillation onset, bistability, stability envelopes, bifurcation loci — becomes accessible to the classical analytical toolkit.

The contralateral archetype is the headline example. Mutual inhibition is a feedback loop with gain $w_{12}w_{21}g_1g_2$, and the winner-take-all transition is the pitchfork at unity loop gain — two lines of algebra after the population reduction. The Hopf in the activator–inhibitor loop is the same condition on the negative-sign-product loop. Sign products around feedback loops classify the bifurcation family (Thomas’s rules at LI&F population scale); loop-gain magnitudes locate the bifurcation in parameter space. The three hypotheses tested numerically all hold (Section 4, Section 5, Section 6), and the framework generalises to series, parallel, and positive-loop archetypes (Section 8).

The population framework is a **companion** to the FCS programme, not a replacement. FCS verifies properties at specific parameter points; the population framework predicts the shape of the parameter region. The cross-validation in Section 7 makes the boundary explicit: at the

symmetric corner the two descriptions agree; in the asymmetric arms the discrete simulator’s spike-timing locks dominate in a way the continuous reduction cannot see. Each scale answers what the other cannot.

11. Acknowledgements

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